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Dissertation evaluation report

Author of the Thesis:

Josef Koumar

Title of the Thesis:

Threat Detection in Network Traffic using Time Series Analysis

Submitted to:

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Overall impression

This dissertation is both very professional and a very good read.

Josef Koumar's dissertation addresses a central problem of contemporary network security: how to detect malicious activity in encrypted traffic when payload inspection is no longer a reliable foundation for monitoring. It is framed clearly and convincingly. The dissertation argues that network metadata can be treated as temporal signals and that threat detection can therefore be reformulated as a time-series-analysis problem. This framing is both technically sound and well aligned with the scientific literature as well as current operational realities in ISP-scale monitoring.

The dissertation is also notably well organized. Rather than presenting isolated studies, it develops a coherent structure around three types of time series derived from network traffic:

- Single flow time series,
- Flow time series, and
- Aggregated time series.

This organization is not merely editorial; it forms the conceptual backbone of the dissertation and allows the author to move systematically from taxonomy and representation to specific detection methods and deployment-oriented systems.

A further merit of the dissertation is that it combines theoretical framing, method design, dataset creation, evaluation, and deployment. The work is grounded in many-many published papers, includes explicit contribution statements, and culminates in practical systems such as NetTiSA, periodicity-based detection pipelines, and the NODS anomaly detection system. As a result, the dissertation reads as a mature and coherent body of doctoral work rather than a collection of disconnected experiments. Again, it is written in an enjoyable manner. Enjoyable even for those not so fond of time-series analysis. Even more so for those like to dig deep into time-series, like myself.





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The timeliness and significance of the dissertation

The topic of the dissertation is highly timely. The thesis correctly identifies that widespread encryption, including TLS 1.3, encrypted DNS, and ECH, has substantially reduced the value of classical deep packet inspection for large-scale threat detection. In this context, methods that rely on observable temporal and statistical properties of traffic are not only academically interesting but increasingly necessary.

The significance of the thesis is shown in various ways. Beside addressing timely problems and providing solutions that are both methodologically correct and practically feasible, it presents real-life, tangible solutions.

The significance of the work is strengthened by its emphasis on deployability in real high-speed environments. The author repeatedly addresses throughput, memory, telemetry size, and operational feasibility, rather than limiting the discussion to offline accuracy results. This is particularly important because many academically attractive methods fail precisely at the point of real deployment; here, deployability is a primary evaluation criterion throughout the dissertation.

The thesis is also significant because it contributes resources of lasting value to the research community. It includes publicly released datasets and code, and the dissertation reports that the author's thesis-related publications had already received several dozens of citations and that the associated datasets had been downloaded well over 20 thousand times. This indicates that the work is not only relevant in principle, but has already begun to influence ongoing research and practice. This work is already appreciated not only in the research lab or has a national impact, but very well-received (and used) by international communities like IEEE ComSoc.

Brief summary of the main contributions

In summary, the dissertation makes two broad types of contribution. First, it provides a systematic framework for constructing and analyzing time series from network traffic, formalizing several traffic-derived time-series types and showing how each can support different detection tasks. Second, it turns this framework into concrete methods for encrypted-traffic classification, periodicity-based threat detection, anomaly detection, and anomaly explanation.

To give a little taste what this means:

The thesis contributes a universal TSA-based feature set and the reduced, deployment-oriented NetTiSA flow; a periodicity-detection approach based on Flow time series and Lomb-Scargle analysis, with applications to multiclass traffic classification, cryptomining detection, and botnet C&C detection; the CESNET-CC25 and CESNET-TimeSeries24 datasets; a forecasting-based anomaly detection system deployed in an ISP network; and an LLM-based layer for explaining detected anomalies to human analysts.

On the strengths of the dissertation

This work has a very consistent focus on the gap between academic performance and operational deployment. The NetTiSA work is an excellent example: the author shows that the reduced feature representation preserves similar discriminative performance while drastically lowering telemetry





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and computational cost. NetTiSA requires 113-byte flow records, compared with much larger competing representations, and its performance and memory behavior are shown to be close to classical IP flow export. The dissertation further states that NetTiSA has already been deployed in the production CESNET2 network with eight probes processing roughly 200,000 flows per second.

Another main strength is the methodological quality of the Flow time series chapter. The author does not merely claim that periodic behavior matters; he develops a concrete mathematical and computational pipeline around the Lomb-Scargle periodogram and significance testing, distinguishes different kinds of periodic behavior, and demonstrates applicability across several tasks. The resulting pipeline is reported to process 3 million flows in 5 seconds, to achieve 91% accuracy and 90% F1-score in multiclass classification, up to 95.19% accuracy and 92.21% F1-score in cryptomining detection, and maximal precision for botnet C&C detection across multiple datasets.

Finally, I really appreciated the dissertation's openness and research maturity. It is grounded in a substantial publication record, the author explicitly documents his contribution to each paper, and the thesis includes datasets and code that enable reproducibility and follow-up work.

This open-science orientation materially increases the scientific value of the dissertation, especially in a field where realistic datasets are difficult to obtain and often remain inaccessible.

On the limitations

The main limitation of the dissertation is its breadth.

It is impressive and, overall, positive; however, it also means that the thesis sometimes reads more like a strong research program built around a common theme than a dissertation driven by one sharply unified methodological centerpiece. The expectations and practices - I know very well - differ in doctoral schools, I just wanted to mention that this is very a broad theme.

The three pillars - single-flow features, periodic Flow time series, and aggregated anomaly detection - are of course clearly related, but they differ enough in technical style that some evaluators may perceive the work as somewhat heterogeneous. Actually, I don't have any real objections, as this setup shows the author's broad view, and capabilities.

There is also a little concern in relation to the anomaly-detection and LLM-based explanation components. These appear somewhat less mature than the strongest core parts of the dissertation. The NODS chapter itself acknowledges that forecasting-based models can generate many false positives and therefore require grouping and filtering before alerts become usable for a SoC team. Likewise, the explainability evaluation relies on manually annotated anomalies from a dataset without predefined anomaly labels, and the thesis itself identifies fine-tuning, prompt design, open-source models, and broader real-world analyst interaction as important future directions. This should be handled in their proper place (as future directions).

Nevertheless, these do not weaken the dissertation fundamentally, but they do indicate that this part of the work remains more exploratory than the strongest deployment-oriented detection chapters.





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Questions to the Author

- 1) The dissertation is built around three traffic-derived representations: Single flow time series, Flow time series, and Aggregated time series. Could you clarify the practical decision rule for choosing among these three in a real deployment - in particular, when would you recommend one representation over the others for new threat classes that were not explicitly studied in the thesis?
- 2) The introduced Flow time series methodology relies on detecting periodic structures in sequences of flow arrivals, and it performs very well for threats such as cryptomining and botnet C&C communication. Could you clarify how robust this approach is when the attacker deliberately introduces jitter, adaptive timing, or mixed benign-background traffic to obscure periodicity, and which part of your pipeline you consider the most vulnerable under such evasive behavior?
- 3) In the periodicity-based part of the thesis, you distinguish Clear and Sinusoidal periodic behaviors and build the detection pipeline around Lomb–Scargle analysis and significance testing for unevenly sampled Flow time series. Could you clarify whether you see these two behavior classes as sufficient for most operationally relevant threats, or whether there are important traffic patterns that fall outside this model and would require extending the periodicity framework?
- 4) A strong theme of the thesis is deployability in ISP-scale networks. NetTiSA as well as the periodicity-detection pipeline are presented as deployment-oriented solutions. Could you clarify what you see as the main remaining bottleneck for broader operational adoption: collector overhead, feature extraction cost, long-term state maintenance, false-positive handling, analyst workflow integration, or others?
- 5) In the NODS anomaly-detection system, the models are chosen partly because they are time- and memory-sensitive enough for deployment, even if this means prioritizing deployability over precision. Could you clarify how you would balance this trade-off in a production SoC setting, especially when the goal is not only to detect outliers but to keep alert volumes manageable for analysts?
- 6) The LLM-based component is positioned as a layer that explains anomalies already detected by a prior anomaly-detection framework, using a zero-shot setup rather than domain-specific fine-tuning. Could you clarify how far you believe this approach can generalize across networks and operational contexts, and what would be the most important next step to move it from a promising proof of concept toward a robust production-grade analyst assistant?





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Overall evaluation summary

Overall, I consider this dissertation to be a strong doctoral contribution.

It addresses a problem of clear practical importance, proposes original and technically credible solutions, supports them with real datasets and deployments, and demonstrates a consistent concern for scalability and operational feasibility. In particular, the combination of conceptual framing, deployable NetTiSA design, periodicity-based flow analysis, and ISP-oriented anomaly detection gives the work both scientific depth and practical relevance.

For these reasons, I recommend accepting Josef Koumar's dissertation. In my judgment, it meets the standard expected of doctoral work and, in several respects, exceeds it through its openness, practical validation, and clear relevance to encrypted-traffic security monitoring. The dissertation represents a substantial and well-executed contribution to the field of network threat detection using time series analysis.

Altogether I accept the thesis and recommend for presentation and defense.

Velence, 31/05/2026

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Pal Varga, PhD

